PalmQ10

User Guide

Version 0.1.3-alpha

Private research release

Urban microclimate simulation for QGIS
Maintained by the PalmQ10 development team

Contents

1 Introduction

PalmQ10 is a QGIS plugin and supporting toolchain that runs urban microclimate simulations. You draw an area of interest (AOI) on a map, fill in a form, and PalmQ10 assembles and runs a chain of atmospheric models, returning maps and vertical sections of temperature, humidity, wind, and thermal comfort.

Running these models directly demands substantial command-line experience and a working knowledge of atmospheric modelling. PalmQ10's purpose is to make that workflow reachable from a map interface instead.

1.1 What PalmQ10 produces

For a given AOI and time window, PalmQ10 generates:

- Horizontal maps at chosen heights above ground — air temperature, apparent temperature, wind speed and direction, and heat-exposure metrics such as hours above 38 °C.
- Vertical slices along user-drawn section lines — temperature, humidity, wind, and mean radiant temperature.
- The underlying PALM NetCDF output, for users who want to do their own analysis.

1.2 The component software

PalmQ10 does not implement the physics itself. It orchestrates four established open-source models, each packaged as an Apptainer container so you do not install them separately.

Component	Role
GEO4PALM	Builds the *static driver* — the description of the ground. Terrain elevation, buildings, streets, pavements, vegetation, and land cover are assembled into the file PALM reads as its surface.
WRF / WPS	The Weather Research and Forecasting model and its preprocessing system. Produces regional weather for your dates, providing the large-scale conditions the city-scale simulation sits inside.
WRF4PALM	Translates WRF output into PALM's *dynamic driver* — the time-varying boundary conditions fed to the domain edges.
PALM	The large-eddy simulation model that performs the actual microclimate simulation at metre scale.
Post-processing	PalmQ10's own stage. Converts raw PALM output into the maps, slices, and comfort indices listed above.

The practical consequence is that a run has four stages, and each depends on the one before. The GUI exposes them as separate buttons so you can re-run a single stage rather than starting over.

Scope of this release

v0.1.3-alpha adds **HPC cluster execution**. Earlier releases ran locally only. You can now send the heavy stages (WRF/WPS, WRF4PALM, PALM) to a SLURM or SGE cluster — see Section 5.

2 Before you start

2.1 Software prerequisites

Installation is covered in the separate *Installer Guide*. In summary, you need Windows 10 or 11 with QGIS (Long Term Release), Ubuntu on WSL2, and 7-Zip. The installer places the plugin, the PalmQ10 folder, a conda environment, and the container images.

2.2 Data access keys

PalmQ10 downloads source data from public providers on your behalf. Several require a free account. On first use, fill in `ACCESS_KEYS_EDIT.txt` in the `palmq10/documentation` folder.

Provider	Needed for	Register at
OpenTopography	ESA/COP30 DEM	`portal.opentopography.org`
Earthdata Login	NASA SRTM DEM	`urs.earthdata.nasa.gov`
Terrascope / ESA	Land cover (WorldCover)	`sso.terrascope.be`
Copernicus CDS	Weather forcing data	`cds.climate.copernicus.eu`
AWS CLI	Canopy height dataset	no account needed currently

You do not need all of them

If you use the default FABDEM terrain source, you need neither OpenTopography nor Earthdata. Terrascope and the Copernicus CDS are required for essentially every run.

Your credentials stay on your machine. Neither the PalmQ10 developers nor the upstream model developers receive them.

2.3 Geographic coverage limits

Building data currently covers Valencia and Lecco only

Building footprints and heights come from a prebuilt dataset shipped with PalmQ10 that covers **Valencia (Spain)** and **Lecco (Italy)**. The `Building source` dropdown shows `eubds`, but this is an extract for those two cities — not a general European database. Running an AOI elsewhere will produce a simulation without realistic buildings. Terrain, land cover, streets, and weather forcing are global; only buildings are restricted.

3 The PalmQ10 interface

Open QGIS and click the PalmQ10 icon in the toolbar. The dialog is organised top to bottom in the order you use it: identify the project, define the area, set the grid, choose data sources, then run the stages.

3.1 Project identity

Field	Example	Meaning
Project	`ciutat_vella_2h`	Short unique identifier. Becomes the folder name and the prefix of every output file. Avoid spaces.
City	`Valencia`	City or municipality of the AOI.
Country	`Spain`	Country of the AOI.

3.2 Time window

Field	Example	Meaning
Start datetime	`2025-08-11 00:00:00`	Local simulation start.
End datetime	`2025-08-11 02:00:00`	Local simulation end. For a 24-hour run, use the same hour on the following day.
Output timestep	`60` minutes	How often results are written. Smaller values produce substantially larger files.

There is also a **spin-up** setting (default 6 hours). Spin-up is simulated time before your analysis window that lets the model settle into a physically consistent state; results during spin-up are discarded. Total simulated time is `spinup_hours + analysis window`, so a 2-hour analysis with 6 hours of spin-up simulates 8 hours.

3.3 Grid and domain

These settings determine both the realism and the cost of the run. They are the most consequential choices in the form.

Field	Default	Meaning
Horizontal resolution	`10` m	Grid spacing. Halving it roughly octuples the computational cost, because the domain grows in two horizontal directions and the timestep must shrink.
Vertical resolution	`5.0` m	Near-ground vertical spacing.
Model top elevation	`500` m ASL	Top of the atmospheric domain, measured **from sea level**, not from the ground. See below.
Vertical stretching	`Off`	**Leave this set to Off.** The control is present in the interface but is **not functional in this release** — see Section 7.
Boundary type	`noncyclic`	Use `noncyclic` for city AOIs.
Simulation buffer	`200` m	Margin around your AOI that reduces the influence of the domain edges on results inside it.
API download buffer	`500` m	Extra margin for downloaded source data.
Available cores	`14`	CPU cores for PALM. On HPC this should match what your cluster allocation provides.
Minimum subdomain points	`16`	Guards against inefficient parallel decompositions. Leave at the default.

Model top elevation is measured from sea level

This is the most common configuration mistake. The value is metres *above sea level*, so it must exceed your terrain elevation plus the atmospheric column you want. For a sea-level city with 300 m of atmosphere, use 300. For a valley at 800 m ASL wanting 400 m of atmosphere, use 1200. Set it too low and PALM will have very few vertical levels above the terrain, making near-surface results unreliable.

3.4 Data sources

Field	Default	Notes
DEM source	`fabdem`	Terrain elevation. See comparison below.
Street source	`osm`	OpenStreetMap, or `manual` for your own data.
Pavement source	`osm`	As above.
Building source	`eubds`	Only active option. Valencia and Lecco only (Section 2.3).
Land-use source	`esa`	ESA WorldCover. Only active option.
Sea-surface temp.	`constant`	Use `constant` unless you have a reason to fetch observed SST.

Choosing a DEM source

Source	Credentials	Characteristics
`fabdem`	None	**Recommended.** Global 30 m, with buildings and forests largely removed — which is what PALM wants, since PalmQ10 adds buildings separately. Downloads in seconds to minutes.
`esa`	OpenTopography	Global 30 m Copernicus DEM. May retain building or vegetation artefacts in places.
`nasa`	Earthdata	SRTM, acquired February 2000, 60°N–56°S. Goes through NASA's processing queue: usually minutes, occasionally more than a day.
`manual`	None	Your own raster. Best option where local LiDAR terrain exists.
`flat`	None	Perfectly flat surface, for debugging and idealised tests.

3.5 Overwrite controls

Two dropdowns control whether previously downloaded source data is re-fetched: **Overwrite static source data** and **Overwrite dynamic source data**.

Use `n` for a normal re-run where you only changed grid resolution or core counts — this reuses downloads and saves considerable time. Use `y` when the AOI extent, the dates, or the source datasets changed.

3.6 Output selection

Field	Meaning
Horizontal output height(s)	Heights above ground for map extraction, e.g. `2.0, 10.0`. 2 m is pedestrian height.
Horizontal analysis outputs	Which maps to generate. At least one required.
Vertical analysis outputs	Slice products generated along your section lines.
Slice spacing	Sampling interval along section lines, in metres.
PALM 3D model outputs	Raw variables PALM writes. Leave at default unless reducing file size.

4 Worked example: a two-hour Valencia simulation

This example runs a short simulation over central Valencia. Two hours is small enough to complete quickly while exercising every stage, which makes it a good first run and a good installation test.

4.1 Step 1 — Create the project

Open QGIS, click the PalmQ10 toolbar icon, and enter:

```
Project : valencia_ciutat_vella_2h
City : Valencia
Country : Spain
```

Click **Open / create project**. PalmQ10 creates the project folder, an AOI GeoPackage, and supporting map layers, and adds them to your QGIS project.

4.2 Step 2 — Draw the area of interest

Click **Draw AOI**. The AOI layer enters edit mode. Draw a polygon over central Valencia — roughly 1 km × 1 km is appropriate for a first run. Click **Draw AOI** again to save and leave edit mode.

Keep the first AOI small

Domain cost grows with area *and* inversely with grid spacing. A 1 km^2 area at 10 m resolution is a reasonable first run. Large areas at fine resolution can produce domains too big to simulate — an oversized domain is a known cause of out-of-memory failures.

4.3 Step 3 — Draw section lines (optional)

If you want vertical slices, click **Draw section lines** and draw one or more lines across the AOI — for instance along a street canyon. Click again to save. Skip this if you only want horizontal maps.

4.4 Step 4 — Fill in the settings

```
Start datetime : 2025-08-11 12:00:00
End datetime : 2025-08-11 14:00:00
Output timestep : 30
Horizontal resolution : 10
Vertical resolution : 5.0
Model top elevation : 300 # Valencia is near sea level
Vertical stretching : Off
```

```
Boundary type : noncyclic
Simulation buffer : 200
API download buffer : 500
Computing backend : local
Available cores : 14 # match your machine
DEM source : fabdem
Horizontal output height: 2.0
```

Click **Save project**. This writes your settings, the AOI geometry, and the section lines, and regenerates the base namelist.

Reduce spin-up for a test run

The 6-hour default means a 2-hour analysis simulates 8 hours. For a first installation test, setting spin-up to 3 hours cuts the runtime substantially. Use the full 6 hours for results you intend to interpret.

4.5 Step 5 — Run the stages

Run these in order, waiting for each to finish:

1. **Prepare static** — sets up GEO4PALM folders and downloads terrain, land cover, streets, and buildings.
2. **Create static** — runs GEO4PALM to build the static driver.
3. **Prepare dynamic** — writes the WRF/WPS and WRF4PALM configuration.
4. **Create dynamic** — runs WRF/WPS and WRF4PALM. *This is normally the longest stage.*
5. **Prepare PALM** — stages the drivers into the PALM run folder.
6. **Run PALM** — runs the simulation.
7. **Run post-processing** — produces the maps and slices.

There is also a **Run full workflow** button that performs all of these in sequence. For a first run, prefer the individual buttons: if something fails you will know exactly which stage it was.

4.6 Step 6 — Look at the results

Post-processing writes into the project folder. Horizontal maps can be loaded as QGIS raster layers; vertical slices are written as images and data files. The raw PALM NetCDF remains available if you want to analyse it yourself.

If the static step fails on land-cover download

The ESA WorldCover download is intermittently unreliable — the provider resets connections under load. This is not a credential problem. Re-run **Prepare static**; it normally succeeds on a retry.

5 Running on an HPC cluster

Local runs are limited by your workstation. For larger domains, longer periods, or finer resolution, PalmQ10 can send the heavy stages to a cluster. This is new in v0.1.3-alpha.

A detailed walkthrough is in the separate *Cluster Connection Manual*. This section covers the concept and the minimum steps.

5.1 How it works

Set **Computing backend** to `HPC` and choose a cluster profile. The static and post-processing stages still run locally; WRF/WPS, WRF4PALM, and PALM run on the cluster. PalmQ10 uploads inputs, submits the job, monitors it, and downloads results.

Cluster profiles are YAML files describing one cluster — hostname, username, scheduler type (SLURM or SGE), storage layout, and paths. Two are shipped, for the UPV Rigel and Sirius clusters. New clusters are added by creating a profile; there is no per-cluster special handling in the code.

5.2 One master password per job

Authenticating to a cluster used to mean typing your cluster password four or five times per job, because each stage reconnects and any brief VPN interruption forced a new login.

PalmQ10 now uses key-based authentication with an encrypted local vault:

- You choose a **master password** once. It is local to your computer and is never stored or transmitted.
- PalmQ10 creates an SSH key for you and registers it with the cluster. You enter your **cluster login password exactly once**, during setup. It is never saved to disk.
- Each job asks for your master password **once**. Everything after that — uploads, submission, monitoring, downloads — proceeds without prompts.
- If your VPN or WiFi drops mid-job, PalmQ10 reconnects silently and resumes interrupted transfers.

There is no master password recovery

This is deliberate — recovery would defeat the protection. If you forget it, re-run cluster setup: PalmQ10 generates a fresh key and re-registers it, and you enter your cluster login password once more.

5.3 First-time cluster setup

The supplied profiles are set up for the maintainer's account

The Rigel and Sirius profiles that ship with PalmQ10 contain the maintainer's cluster username. If you are a different user, you **must** edit the profile before your first job — and the username appears in more than one place. Change `username` *and every remote path that contains it*: `remote_project_root`, `remote_scratch_root`, `remote_geog_dir`, `remote_container_dir`, `remote_assets_root`, and each entry under `remote_images`. Changing only `username` leaves the paths pointing at someone else's home and scratch directories, and the job fails with permission errors that do not obviously point back to the profile.

1. Open the cluster profile dialog and select or create a profile.
2. **Set your username in the profile, and update every remote path that contains a username** — see the note above. On some systems the username includes a domain, e.g. `yourname@upvnet.upv.es`.
3. Use **Set / change HPC master password** to create your master password.
4. Click **Check HPC assets**. This verifies whether the container images and WPS_GEOG data are present on the cluster.

5. If anything is missing, use the upload buttons. **These transfers are large** — the geographic dataset in particular. Start them when you can leave the machine running; they resume automatically after network interruptions.

6. Set **Computing backend** to `HPC`, select the profile, and run as normal.

6 Troubleshooting

Symptom	Likely cause and fix
Static step fails downloading land cover	Provider connection reset. Re-run the stage; it usually succeeds on retry.
Simulation has no buildings	Your AOI is outside Valencia or Lecco (Section 2.3).
PALM stops with a grid or building error	Usually the model top is too low, or vertical stretching begins below roof height. Verify model top elevation is measured from sea level and exceeds terrain plus your intended atmospheric column.
Run fails with out-of-memory	Domain too large. Reduce AOI area or coarsen horizontal resolution.
Asked for a cluster password mid-job	Key authentication is not active for that profile. Re-run cluster setup.
Thermal comfort output is empty	Check that the required radiation variables are included in the PALM 3D outputs.

7 Known limitations in this release

- **Buildings cover Valencia and Lecco only.** See Section 2.3.
- **Vertical stretching is not functional.** The *Vertical stretching* control appears in the interface, but the feature is not yet implemented in this release. Leave it set to `Off`; it is planned for a future version.
- **Windows only.** Windows 10/11 with QGIS and Ubuntu on WSL2. There is no macOS or native Linux installer.
- **Lake physics is not enabled.** Water bodies are represented at a fixed temperature. Lake modelling is planned.
- **Alpha status.** Some steps require user interaction and some failures surface as raw model errors.
- **Cluster support is validated on two clusters** (UPV Rigel and Sirius). Other SLURM or SGE systems should work but are untested.

8 Further documentation

- *Installer Guide* — installation and first-use setup.
- *Cluster Connection Manual* — detailed HPC configuration.
- *Technical Guide* — architecture and internals, for developers.

- *What's New in v0.1.3-alpha* — changes since the previous release.

Licensing. Underlying components — notably PALM, WRF, GEO4PALM, and WRF4PALM — carry their own licence terms. Consult each project for conditions governing use and publication.